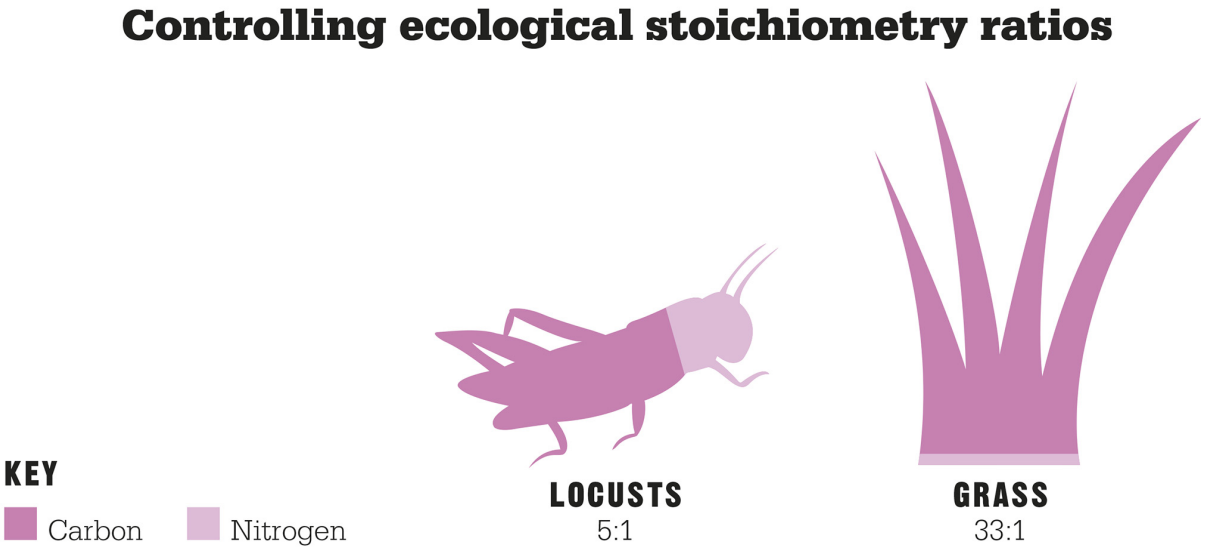
off, breathing out the excess carbon as CO_2. Overuse of such mechanisms to redress a high imbalance can, however, affect fitness, growth, and reproduction. An animal that eats other animals has less work to do, because its prey's C:N:P ratio closely matches its own. However, the size of its prey population is still determined by the plants in its environment because plants with a high carbon ratio can only support a small food chain of consumers.



A locust eats grass that may contain six times as much carbon as it needs. To get the right balance, it excretes carbon or breathes it out as CO_2. Locusts are widely used in research because they are easy to breed.

Understanding our world

Food chains are one area of study; ecological stoichiometry covers just about everything and all the links in between. By discovering how the chemical content of organisms shapes their ecology, scientists are also learning how environments can be better managed. Their findings may significantly influence the future of life on Earth.